



RB003 Week 4 — English Worksheet

Topic: Dictionaries — Items with Keys and Values · **Course:** Tycoon Engineer · **Time:** about 45 minutes

This week one item holds **many facts at once** using a **dictionary** — keys like `Name` and `Price` with values after the `=`. You read a fact with a **dot**: `item.Price`.

1 · Predict 🧠

Read each script. Before you imagine running it in Roblox Studio, write what you think prints in the Output window.

```
local item = {Name = "Gold Ore", Price = 50, Payout = 12}
print(item.Name)
print(item.Price)
```

What two lines print?


```
local item = {Name = "Gold Ore", Price = 50, Payout = 12}
print(item.price)
```

Careful with capital letters! The key is `Price` but the script asks for `price`. What prints, and why?


```
local catalog = {  
    {Name = "Fast Dropper", Price = 100},  
    {Name = "Gold Ore", Price = 250},  
}  
  
for i, item in ipairs(catalog) do  
    print(i, item.Name, item.Price)  
end
```

How many lines print? What exactly is on the second line?

2 · Spot the Bug 🐛

Each block of code below was meant to do something, but it is broken. Read what the code is **supposed** to do, then rewrite it so it works. After that, explain why the original was wrong and why your fix works.

Bug A — The script should print the price, `50`. Instead it prints `nil`.

```
local item = {Name = "Gold Ore", Price = 50}
print(item.price)
```

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — The script should make one item with a name and a price. Instead, Roblox Studio shows a **red error** before the game even runs.

```
local item = {Name "Gold Ore", Price = 50}
print(item.Name)
```

Hint: every key needs an `=` before its value.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — The script should print the item's name, `Gold Ore`. The coder used `[1]` like an array, but a dictionary has **keys**, not numbered slots — so it prints `nil`.

```
local item = {Name = "Gold Ore", Price = 50}
print(item[1])
```

Write the fixed code:

Why was it wrong? Why does your fix work?

3 · Tell Me What You Built 📷

Now switch to Roblox Studio and open your tycoon. Build a `catalog` — a list of at least 2 item dictionaries, each with a `Name` and a `Price` — and loop through it with `ipairs` to print every item's name and price. When you finish, come back here.

Send a photo or video of your script and the printed catalog, then explain what you did. Use these sentence starters — write 4 to 6 sentences total.

First, I ...

My dictionary stores ...

I used a dot to ...

One tricky moment was when ...

To fix it, I ...

If I had more time, I would ...

4 · Record Your Walkthrough 🎥

Now take a video on your phone (or a parent's phone) while you run your script. Talk through it like you are teaching someone who has never seen it. Try to use these words: **dictionary**, **key**, **value**, **dot**, **catalog**.

1. Show your `catalog` in the script.
2. Press Play and show every name and price printed in the Output window.
3. Read one dictionary out loud — say its keys and its values.
4. Say in your own words why `item.price` (small p) gives `nil` .

Write what you will say in your video. Use the space below to plan it before you record — you can read from it while filming.

Submit ✓

Send this worksheet + your walkthrough video to teacher on KakaoTalk.